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Herfindahl–Hirschman index level of concentration values modification and analysis of their change

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Abstract The Herfindahl–Hirschman Index (*HHI*) that measures the level of concentration in a given industry is a well-known and commonly accepted one indicator of market competition. On the basis of European Union Commission guidelines and *HHI* values, the given industry can be characterized as unconcentrated, moderately concentrated or concentrated. The paper is given over to a sensitivity analysis of *HHI* values, which allows simulations of concentration changes in relevant markets in order to assess new entries to the market. This paper derives relationships that allow the setting of boundaries in which the characteristic of industry concentration remains the same. Derived relations of *HHI* sensitivity analyses can be successfully used as a tool in assessing the entry of new subjects into any industry. In the case of economies with a smaller number of operating undertakings, it may be difficult to use the analysis based on the methodology of the European Commission. Therefore, the authors propose an approach of setting boundary ranges to characterize the concentration of the industry. Also, empirical analysis of the Slovak insurance industry, in which 23 insurance companies currently operate, was conducted on the basis of the European Commission methodology.

Keywords Herfindahl–Hirschman Index · Industry Concentration · Relevant Market · Sensitivity Analysis · Slovak Insurance Market

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1 Introduction

An important part of the economic policy of each country is to stimulate and maintain effective competition.[1] and a competitive environment[2] Competitive structure in an industry can have different characteristics depending on the number and size of entities operating. One of the possible forms of competitive structures of industry is concentrated industry. Based on the relationship between the scope of the undertakings of the businesses concerned, concentration can take various forms: horizontal concentration and non-horizontal (vertical and conglomerate) concentration. In the case of horizontal concentration, the undertakings concerned are actual or potential competitors in the same relevant market and thus producing the same or similar products and selling them under the same geographical conditions. The result of horizontal concentration is a strengthening of the dominance of enterprises and a reduction in the number of firms operating in the given market. In the case of vertical concentration, the firms concerned are operating in different, but relevant, markets—with, for example, different levels of the supply chain, when a manufacturer merges with one of its distributors. Based on the direction of mergers in a supply chain, we can distinguish downstream and upstream integration. This leads to an intensification in supply chain relations resulting in a higher concentration level. Conglomerate concentration is a result of mergers between firms that are in a relationship which is neither purely horizontal nor vertical. In this case, the firms concerned are active in closely-related markets. In this paper, we focus on an analysis of horizontal concentration because the evolution of horizontal concentration in the industry and its level is a key factor characterizing the level of the competitive environment. A second reason relates to how its exact definition is based on a system of quantitative characteristics and how these quantitative characteristics are an important part of the analytical materials of state institutions involved in monitoring compliance with competition rules in most developed economies.

A considerable amount of literature (Brezina 1994; Fendeková and Fendek 1997; Hannah and Kay 1977; Leach 1992) has been published on the evaluation methods of concentration consequences in imperfectly competitive markets. All of them are essentially based on market share. Market share can be defined as the proportion of an industry's or market's total sales earned by a particular company in a given market area over a specified time period. Let n represent the number of entities operating in a given market or industry and q_i represent sales volume of an i-th entity operating in a given market or industry ($i = 1, 2, \ldots, n$), then the market share (s_i) of the i-th entity operating on given market or industry can be defined as:[3]

[1] The term 'competition' is considered to be a set of certain conduct rules for business and the state acting on a given market, in which compliance is a prerequisite for further economic development (Decree No. 204/2009 of the Antimonopoly Office of the Slovak Republic 2009).

[2] The term 'competitive environment' is understand to be the environment within which several subjects operate and none of the actors has a supply power large enough that it somehow controls and sets other players at a disadvantage (Fendeková and Fendek 1997).

[3] Market share may have values $0 < s_i = 1$. The sum of market shares of all entities operating on given market is equal to one.

$$s_i = \frac{q_i}{\sum_{i=1}^{n} q_i} i = 1, 2, \ldots, n.$$

All of the following indicators are based on market share. Depending on whether the indicators quantify the concentration level in respect of all subjects, or only in respect to their particular feature subset, we can divide the indicators into two groups e.g. (Fendeková and Fendek 1997):

1. Absolute concentration measure,
2. Relative concentration measure.

Absolute and relative concentration measures significantly differ in the way they take into account the number of entities that carry the characteristic of interest. Next, we will discuss only the indicators for measuring absolute concentration.

Among the well-known quantitative approaches used to analyze markets are statistical and econometric methods, game theory, and special indices (Brezina 1994). Analysis of the concentration of any industry can be carried out through indices, which allow us to analyze the condition of the competitive behavior of entities operating in the relevant market. Such analysis provides regulatory authorities with information about the structure of the market and allows them to lay down recommendations for specific economic action.

Among the most important indices for measuring the concentration of the industry is the Concentration ratio of m the most powerful entities in the industry denoted by CR_m [the US Department of Justice has used it since 1968 (Merger Guidelines 1968)] and the Herfindahl–Hirschman index of market concentration denoted by *HHI* [used by the US Department of Justice since 1982 (Merger Guidelines 1982)]. The Gini index, denoted by *GI*, is also relatively quite often used and is based on the Lorenz curve capturing the actual allocation of market shares.

In assessing the condition of the industry competitive environment, other special indices may also be used, e.g. the Comprehensive concentration index *CCI* (Horvat 1970), Rosenbluth index *RI* (Rosenbluth 1955), Hannah-Kay index *HKI* (Hannah and Kay 1977) etc. Many of these analyses combine different indices, e.g. (Fedderke and Szalontai 2005) measured South Africa production industry concentration based on the Herfindahl–Hirschman index in combination with the Rosenbluth and Gini indices. Among other authors who have combined different indices may also be included for example: (Kwoka 1977; Leach 1992; Golan et al. 1996; Bikker and Haaf 2002; Bajo and Salas 2002; Mare 2005; Naude 2006).

This paper presents the possibility of extending the application of Herfindahl–Hirschman index, the most widely used index for measuring absolute concentration in an industry, which is mentioned in many countries' guidelines for assessing mergers of undertakings.

The main part of this paper is focused on a brief description of the nature of the Herfindahl–Hirschman index and its boundaries that are currently used for measuring the level of concentration of industry on the basis of existing legislation in the US and in the European Union (EU). Following that description, we present proposals to amend the boundaries which should be applied in cases of relevant markets with a smaller number of operating entities.

In terms of industry concentration measurements, it is interesting to see how new entries into the industry change its characteristics. Generally, for these cases, two basic situations can be considered. In the first case, we can assume that the given market or industry is able to absorb additional production volume of the new entity, namely the total production volume of the industry will increase after the entry of a new entity It is assumed production volume by others in the industry remain the same. In a second case, it can be assumed that the industry is not able to absorb additional production volume of the new entity, namely the total production volume of the industry remains the same after the entry of the new entity. The volume of production of other entities will be reduced in proportion to the original shares.

The authors of this paper have focused on the derivation of relations allowing a sensitivity analysis of the *HHI* index based on these two assumptions and derive formulas for determining intervals in which industry characteristics of concentration won't change under the current legislation of the European Commission. These assumptions have been analyzed after being utilized in an empirical analysis of the Slovak insurance market. The Slovak insurance industry represents an important part of the Slovak financial market and, through various indicators, tells us not only about the state of the insurance industry in the country, but also about the state of the economy. The development of the Slovak insurance market essentially correlates with the development of the economy. According to the current state of development and its indicators, the Slovak insurance market is approaching the level of developed insurance markets of European Union (Analysis of the Slovak Financial Sector for 2012 2013).

The structure of the paper is divided into the following interrelated parts. In the introduction, the authors present an overview of the literature devoted to the measurement of absolute concentrations, primarily to special indices used for description of characteristics of a relevant market. The second part is devoted to the most commonly used Herfindahl–Hirschman index (HHI). The third part presents some common extensions of HHI. Core part of the paper is devoted to the modification of HHI intervals for relevant markets with a smaller number of operating entities and the sensitivity analysis of HHI (fourth part), together with the analytical derivation of corresponding intervals in the case of new entry into the relevant market (fifth part). The sixth part is the link of assumptions for modified HHI intervals from the fourth part and the sensitivity analysis presented in the fifth part. Verification of presented considerations is implemented in the last part of paper, devoted to the empirical analysis of concentration in the Slovak insurance market.

2 Herfindahl–Hirschman index

As mentioned above, the Herfindahl–Hirschman concentration index (denoted by *HHI*) is among the most widely used indices for measuring absolute concentration in industry, and it has been part of the US directives on horizontal mergers since 1982, and as the most widely used concentration level measure, *HHI* is a constituent part of guidelines of other countries for assessing mergers of undertakings. *HHI* was independently developed by (Hirschman 1945) and (Herfindah 1950). *HHI* is a convex function of the market shares of all subjects in a given sector (in a relevant market). *HHI* can be defined as the sum of the squares of the market shares s_i $(i = 1, 2, \ldots, n)$ of all entities in the industry:

$$HHI = \sum_{i=1}^{n} (s_i)^2 \tag{1}$$

Let us consider the case where only a single entity operates in a given market and controls the entire industry supply ($s_1 = 1$), then the *HHI* value is equal to one ($HHI = (1)^2 = 1$).

Vice-versa, if we consider the case where n entities operate in a given market with equal market shares $s_i = 1/n, \quad i = 1, 2, \ldots, n$, then *HHI* is expressed by the following formula:

$$HHI = \sum_{i=1}^{n} \left(\frac{1}{n}\right)^2, \quad i = 1, 2, \ldots, n. \tag{2}$$

Because in this case, the market shares of all entities are the same, the following applies:

$$HHI = n \left(\frac{1}{n}\right)^2 = \frac{1}{n}. \tag{3}$$

Thus, *HHI* can result in two extreme values, a maximum value of one (in the case where a market supply is represented by a single operating entity) or a minimal value $1/n$ (in the case where all entities have equal market share).

HHI is primarily used when assessing the level and consequences of concentration in imperfectly competitive markets. According to the calculated value of the *HHI* index, the U.S. Federal Trade Commission (FTC) classifies markets into three types (Horizontal Merger Guidelines 2010):

Unconcentrated,	if the value of *HHI* is less than 0.15,
Moderately concentrated,	if the value of *HHI* is in range $\langle 0.15; 0.25 \rangle$,
Highly concentrated,	if the value of *HHI* is greater than 0.25.

In the assessment of horizontal mergers, it is considered whether the change in concentration after the merger is more than 1 % (i.e. $\Delta HHI = 0.01$) in cases of moderately concentrated markets, and by more than 0.5 % (i.e. $\Delta HHI = 0.005$) in cases of highly concentrated markets. Such a merger leading to increased concentration in the industry is very carefully considered.

Our adopted classification of concentration levels in industry is the one used by the European Commission, according to the Guidelines on the assessment of horizontal mergers under the Council Regulations on the control of concentrations between undertakings (Official Journal of the European Union 2004):[4]

[4] Special circumstances are listed in the Guidelines on horizontal mergers assessed under the Council Regulation on the control of concentrations between undertakings (Official Journal of the European Union 2004).

Unconcentrated,	if the value of *HHI* is less than 0.1,
Moderately concentrated,	if the value of *HHI* is in range $\langle 0.1; 0.2\rangle$ and ΔHHI is less than 0.025,
Highly concentrated	if the value of *HHI* is greater than 0.2 and ΔHHI is less than 0.015 except where there are special circumstances.

3 Extension of the Herfindahl–Hirschman index

A modified, extended *HHI* is called the Normalized Herfindahl–Hirshman Index (*NHHI*). Unlike *HHI,* which ranges between $\langle 1/n, 1\rangle$, the *NHHI* ranges between values from an interval of $\langle 0, 1\rangle$ and is calculated as follows (Khurshid et al. 2009):

$$NHHI = \frac{(HHI - 1/n)}{1 - 1/n} \tag{4}$$

Another modification of *HHI* is represented by the Hannah–Kay index (Hannah and Kay 1977):

$$HK_\alpha = \begin{cases} \left(\sum_{i=1}^{n} s_i^{\alpha}\right)^{1/(\alpha-1)} & \text{if } \alpha > 0, \alpha \neq 1 \\ \prod_{i=1}^{n} s_i^{s_i} & \text{if } \alpha = 1 \end{cases} \tag{5}$$

where α is the elasticity parameter that determines the weight attributed to large entities with respect to small entities. The value of HK_α is equal to the value of *HHI* if $\alpha = 2$.

Ten Kate (Kate 2006) discloses the parallel use of the Dominance Index (*DI*) proposed by Garcia Alba (Iduñate 1990), together with *HHI*. When assessing the merger of two or more entities, the main difference between *HHI* and *DI* is the fact that, while the value of *HHI* increases, the value of *DI* does not necessarily increase. In particular, if the two entities have a combined market share of less than the dominant operator, the *DI* is declining, whereas the *HHI* would increase. If all entities operating on the market are equal, the *DI* coincides with the *HHI* index. The definition of *DI* is based on the *HHI*. Instead of direct market share s_i, it is calculated with relative contributions σ_i of individual entities relative to the original *HHI*:

$$DI = \sum_{i=1}^{n} \sigma_i^2, \text{ where } \sigma_i = \frac{(s_i)^2}{HHI} = \frac{(s_i)^2}{\sum_{i=1}^{n} (s_i)^2} \tag{6}$$

Like *HHI*, the index of dominance may acquire a maximum value of one, when the whole supply is concentrated by a single entity and a minimum value $1/n$ if all entities have the same size.

4 Modification of intervals for Herfindahl–Hirschman index in cases of relevant markets with a smaller number of operating entities

The application of the Herfindahl–Hirschman index and its modification for industry concentration analysis may be problematic in cases of markets, where fewer entities operate in the relevant market. As stated above, in theoretical cases, when all entities operating in a given market have equal market share, *HHI* is equal to $1/n$. Thus the smaller number of entities operating in the market, cause an increase in the value of the lower boundary $1/n$. At the number of ten or fewer entities operating in the given market, it is not possible to achieve the level of concentration characterized as 'unconcentrated'. Applied to cases when fewer than six entities operate in the market, the sector has to be characterized as 'concentrated'.

The determination of interval boundaries for assessing the level of concentration in the industry can thus be derived just from the lower bound value $1/n$. The values used by the European Commission represent the first decile ($Q_{1/10}$) and second decile ($Q_{2/10}$) of the considered interval values. Accepting intervals for the distribution $\langle 1/n, 1\rangle$ in deciles, the value $0.9/n + 0.1$ limits the interval value of the first decile, resp. $0.8/n + 0.2$ limits the interval values of the second decile. With such reasoning it should be possible to modify the classification of level concentration in an industry, based on *HHI* given the number of entities operating in the market *n*, as follows:

$$\begin{array}{ll}
\textit{Unconcentrated}, & \text{if the value of } HHI \text{ has an interval } \left\langle \frac{1}{n};\ \frac{0.9}{n} + 0.1 \right),\\
\textit{Moderately concentrated}, & \text{if the value of } HHI \text{ has an interval } \left\langle \frac{0.9}{n} + 0.1;\ \frac{0.8}{n} + 0.2 \right\rangle,\\
\textit{Highly concentrated}, & \text{if the value of } HHI \text{ has an interval } \left(\frac{0.8}{n} + 0.2;\quad 1 \right\rangle.
\end{array}$$

According to the classification of the level of concentration in the industry used by the EU, an industry is considered to concentrated a value of *HHI* greater than 0.2, which on the basis of Eq. (3) is the case with at least five entities and according to the classification of FTC, the value *HHI* is greater than 0.25, which on the basis of that relationship (3) covers least six entities. According to the modified classification of concentration level in the industry, it is relevant to consider only cases when $n \geq 6$.

5 Sensitivity analysis of Herfindahl–Hirschman index values

As mentioned above, to calculate the *HHI*, Eq. (2) is applied. In cases of new entrants into the industry, it is interesting to see at what value of the new entrant' market share the characteristics of the given industry are changed. A sensitivity analysis of the *HHI* calculated values, given the new input data, can be used for this.

Based on the following, two basic assumptions for *HHI* values sensitivity analysis were carried out:

I. Given that the market or industry is able to absorb additional production volume of a new entrant ΔQ, i.e. the total production volume of the industry will increase to $Q + \Delta Q$ after the entry of a new entity. It is assumed production volume by others in the industry remain the same.
II. Given that the market or industry is not able to absorb additional production volume of the new entity ΔQ, i.e. the total production volume of the industry remains the same after the entry of a new entity, equal to Q. The volume of production of other entities shall be reduced in proportion to the original shares.

5.1 The total industry production volume is increased after the entry of a new entity

The interval of values in which the production volume of new entrants can vary without impacting on the concentration characteristics of given industry, are going to be derived based on Eq. (7). Expression $\frac{\Delta Q}{Q+\Delta Q}$ represents the market share of the new entity on increased production volume of the given industry and $\frac{q_i}{Q+\Delta Q}$ represents the proportion of the original i-th entity on production volume after the entry of the new entity into the industry. Then *HHI* is calculated according to the formula:

$$HHI = \left(\frac{\Delta Q}{Q + \Delta Q}\right)^2 + \sum_{i=1}^{n} \left(\frac{q_i}{Q + \Delta Q}\right)^2. \tag{7}$$

From Eq. (7) can be expressed $\Delta Q[\Delta Q_1, \Delta Q_2$ represent solutions of Eq. (7)]:

$$\Delta Q_{1,2} = \frac{HHI2Q \pm \sqrt{(-HHI2Q)^2 - 4\,(1 - HHI)\left(\sum_{i=1}^{n} q_i^2 - HHIQ^2\right)}}{2\,(1 - HHI)}. \tag{8}$$

From Eq. (8), the intervals for the level of concentration can be derived according to the EU classification. On the basis of these values, it is possible to determine the intervals in which the new entrant's production volume can range after its entry into the industry, so that the industry is considered 'unconcentrated', 'moderately concentrated' or 'concentrated'.

The intervals of the level of concentration according to EU classification can be calculated based on these equations:[5]

$$\Delta Q_{1,2}^{HHI=0.1} = \frac{0.2Q \pm \sqrt{0.4Q^2 - 3.6\sum_{i=1}^{n} q_i^2}}{1.8}, \tag{9}$$

Let the expression under the square root in (9) be denoted as D_1,

$$\Delta Q_{1,2}^{HHI=0.2} = \frac{0.4Q \pm \sqrt{0.8Q^2 - 3.2\sum_{i=1}^{n} q_i^2}}{1.6}, \tag{10}$$

and let the expression under the square root in (10) be denoted as D_2.

From Eqs. (9) and (10) it is possible to derive intervals of ΔQ, so the industry is considered as 'unconcentrated' after the entry (let the production volume of the new entity in the case of 'unconcentrated' industry be denoted as ΔQ^U); when applied, we get the following:

[5] When applying the FTC classification in Eq. (8) the values of *HHI* = 0.15 and *HHI* = 0.25 are used (Horizontal Merger Guidelines 2010).

$$\Delta Q^U \in \begin{cases} \varnothing; & \text{if } D_1 \leq 0 \\ \left(\frac{0.2Q-\sqrt{D_1}}{1.8};\ \frac{0.2Q+\sqrt{D_1}}{1.8}\right); & \text{if } D_1 > 0,\ \sqrt{D_1} < 0.2Q \\ \left(0;\ \frac{0.2Q+\sqrt{D_1}}{1.8}\right); & \text{if } D_1 > 0,\ \sqrt{D_1} \geq 0.2Q \end{cases} \tag{11}$$

The industry is considered to be 'moderately concentrated' after its entry (let the production volume of the new entity in the case of 'moderately concentrated' industry be denoted as ΔQ^{MC}), if applied, we get the following:

$$\Delta Q^{MC} \in \begin{cases} \varnothing; & \text{if } D_2 < 0 \\ \frac{0.4}{1.6}Q; & \text{if } D_2 = 0 \\ \left\langle\frac{0.4Q-\sqrt{D_2}}{1.6};\ \frac{0.4Q+\sqrt{D_2}}{1.6}\right\rangle; & \text{if } D_2 > 0,\ D_1 \leq 0,\ \sqrt{D_2} < 0.4Q \\ \left\langle 0;\ \frac{0.4Q+\sqrt{D_2}}{1.6}\right\rangle; & \text{if } D_2 > 0,\ D_1 \leq 0,\ \sqrt{D_2} \geq 0.4Q \\ \left\langle\frac{0.4Q-\sqrt{D_2}}{1.6};\ \frac{0.2Q-\sqrt{D_1}}{1.8}\right\rangle & \\ \quad \cup \left\langle\frac{0.2Q+\sqrt{D_1}}{1.8};\ \frac{0.4Q+\sqrt{D_2}}{1.6}\right\rangle; & \text{if } D_2 > 0,\ D_1 > 0,\ \sqrt{D_2} < 0.4Q \\ \left\langle 0;\ \frac{0.2Q-\sqrt{D_1}}{1.8}\right\rangle & \\ \quad \cup \left\langle\frac{0.2Q+\sqrt{D_1}}{1.8};\ \frac{0.4Q+\sqrt{D_2}}{1.6}\right\rangle; & \begin{array}{l}\text{if } D_2 > 0,\ D_1 > 0, \\ \sqrt{D_2} \geq 0.4Q,\ \sqrt{D_1} < 0.2Q\end{array} \\ \left\langle\frac{0.2Q+\sqrt{D_1}}{1.8};\ \frac{0.4Q+\sqrt{D_2}}{1.6}\right\rangle; & \begin{array}{l}\text{if } D_2 > 0,\ D_1 > 0, \\ \sqrt{D_2} \geq 0.4Q,\ \sqrt{D_1} \geq 0.2Q\end{array} \end{cases} \tag{12}$$

The industry is considered to be 'concentrated' after its entry (let the production volume of the new entity in the case of 'concentrated' industry be denoted as ΔQ^C), if applied, we get the following:

$$\Delta Q^C \in \begin{cases} (0; \infty); & \text{if } D_2 < 0 \\ \left(0;\ \frac{0.4Q-\sqrt{D_2}}{1.6}\right) \cup \left(\frac{0.4Q+\sqrt{D_2}}{1.6}; \infty\right); & \text{if } D_2 \geq 0,\ \sqrt{D_2} < 0.4Q \\ \left(\frac{0.4Q+\sqrt{D_2}}{1.6}; \infty\right); & \text{if } D_2 \geq 0,\ \sqrt{D_2} \geq 0.4Q \end{cases} \tag{13}$$

5.2 The total industry production volume is unchanged after the entry of a new entity

The interval of values in which the production volume of new entrants can vary without impacting on the concentration characteristics of a given industry, reflecting the fact that the industry is not able to absorb additional production volume (the volume of production is unchanged), is going to be derived based on Eq. (14). Expression $\frac{\Delta Q}{Q}$ represents the market share of the new entity on the total unchanged production volume of the given industry and $\frac{q_i\left(1-\frac{\Delta Q}{Q}\right)}{Q}$ represents the proportion of i-th entity on production volume after the entry of the new entity into the industry. Then *HHI* is calculated according to the formula:

$$HHI = \left(\frac{\Delta Q}{Q}\right)^2 + \sum_{i=1}^{n}\left(\frac{q_i\left(1-\frac{\Delta Q}{Q}\right)}{Q}\right)^2. \tag{14}$$

From Eq. (14) can be expressed $\Delta Q[\Delta Q_1, \Delta Q_2$ represent solutions of Eq. (14)]:

$$\Delta Q_{1,2} = \frac{2Q\sum_{i=1}^{n} q_i^2 \pm \sqrt{\left(-2Q\sum_{i=1}^{n} q_i^2\right)^2 - 4\left(Q^2+\sum_{i=1}^{n} q_i^2\right)\left(Q^2\sum_{i=1}^{n} q_i^2 - HHIQ^4\right)}}{2\left(Q^2+\sum_{i=1}^{n} q_i^2\right)}. \tag{15}$$

From Eq. (15), the boundary values of ΔQ can be calculated, from which intervals for ΔQ can be derived according to the EU classification. On the basis of these values, it is possible to determine the interval in which the new entrant's production volume can range after its entry into the industry, so that the industry is considered 'unconcentrated', 'moderately concentrated' or 'concentrated'.

The intervals of the level of concentration according to the EU classification can be calculated based on these equations:[6]

$$\Delta Q_{1,2}^{HHI=0.1} = \frac{2Q\sum_{i=1}^{n} q_i^2 \pm \sqrt{\left(-2Q\sum_{i=1}^{n} q_i^2\right)^2 - 4\left(Q^2+\sum_{i=1}^{n} q_i^2\right)\left(Q^2\sum_{i=1}^{n} q_i^2 - 0.1Q^4\right)}}{2\left(Q^2+\sum_{i=1}^{n} q_i^2\right)}, \tag{16}$$

Let the expression under the square root in (16) be denoted as D_3,

$$\Delta Q_{1,2}^{HHI=0.2} = \frac{2Q\sum_{i=1}^{n} q_i^2 \pm \sqrt{\left(-2Q\sum_{i=1}^{n} q_i^2\right)^2 - 4\left(Q^2+\sum_{i=1}^{n} q_i^2\right)\left(Q^2\sum_{i=1}^{n} q_i^2 - 0.2Q^4\right)}}{2\left(Q^2+\sum_{i=1}^{n} q_i^2\right)}, \tag{17}$$

and let the expression under the square root in (17) be denoted as D_4.

The industry is considered to be 'unconcentrated' after its entry (let the production volume of the new entity in the case of 'unconcentrated' industry be denoted as ΔQ^U), if applied, we get the following:

6 When applying FTC classification in Eq. (8) the following values of *HHI* $= 0.15$ and *HHI* $= 0.25$ are used.

$$\Delta Q^U \in \begin{cases} \varnothing; & \text{if } D_3 \le 0 \\ \left(\frac{2Q\sum_{i=1}^n q_i^2-\sqrt{D_3}}{2\left(Q^2+\sum_{i=1}^n q_i^2\right)}; \frac{2Q\sum_{i=1}^n q_i^2+\sqrt{D_3}}{2\left(Q^2+\sum_{i=1}^n q_i^2\right)}\right); & \text{if } D_3 > 0, \sqrt{D_3} < 2Q\sum_{i=1}^n q_i^2 \\ \left(0; \frac{2Q\sum_{i=1}^n q_i^2+\sqrt{D_3}}{2\left(Q^2+\sum_{i=1}^n q_i^2\right)}\right); & \text{if } D_3 > 0, \sqrt{D_3} \ge 2Q\sum_{i=1}^n q_i^2 \end{cases} \tag{18}$$

The industry is considered to be 'moderately concentrated' after its entry (let the production volume of the new entity in the case of 'moderately concentrated' industry be denoted as ΔQ^{MC}), if applied, we get the following:

$$\Delta Q^{MC} \in \begin{cases} \varnothing; & \text{if } D_4 < 0 \\ \frac{2Q\sum_{i=1}^n q_i^2}{2\left(Q^2+\sum_{i=1}^n q_i^2\right)}; & \text{if } D_4 = 0 \\ \left\langle\frac{2Q\sum_{i=1}^n q_i^2-\sqrt{D_4}}{2\left(Q^2+\sum_{i=1}^n q_i^2\right)}; \frac{2Q\sum_{i=1}^n q_i^2+\sqrt{D_4}}{2\left(Q^2+\sum_{i=1}^n q_i^2\right)}\right\rangle; & \begin{array}{l}\text{if } D_4 > 0,\, D_3 \le 0, \\ \sqrt{D_4} < 2Q\sum_{i=1}^n q_i^2\end{array} \\ \left\langle 0; \frac{2Q\sum_{i=1}^n q_i^2+\sqrt{D_4}}{2\left(Q^2+\sum_{i=1}^n q_i^2\right)}\right\rangle; & \begin{array}{l}\text{if } D_4 > 0,\, D_3 \le 0, \\ \sqrt{D_4} \ge 2Q\sum_{i=1}^n q_i^2\end{array} \\ \left\langle\frac{2Q\sum_{i=1}^n q_i^2-\sqrt{D_4}}{2\left(Q^2+\sum_{i=1}^n q_i^2\right)}; \frac{2Q\sum_{i=1}^n q_i^2-\sqrt{D_3}}{2\left(Q^2+\sum_{i=1}^n q_i^2\right)}\right\rangle \\ \quad \cup \left\langle\frac{2Q\sum_{i=1}^n q_i^2+\sqrt{D_3}}{2\left(Q^2+\sum_{i=1}^n q_i^2\right)}; \frac{2Q\sum_{i=1}^n q_i^2+\sqrt{D_4}}{2\left(Q^2+\sum_{i=1}^n q_i^2\right)}\right\rangle; & \begin{array}{l}\text{if } D_4 > 0,\, D_3 > 0, \\ \sqrt{D_4} < 2Q\sum_{i=1}^n q_i^2\end{array} \\ \left\langle 0; \frac{2Q\sum_{i=1}^n q_i^2-\sqrt{D_3}}{2\left(Q^2+\sum_{i=1}^n q_i^2\right)}\right\rangle \\ \quad \cup \left\langle\frac{2Q\sum_{i=1}^n q_i^2+\sqrt{D_3}}{2\left(Q^2+\sum_{i=1}^n q_i^2\right)}; \frac{2Q\sum_{i=1}^n q_i^2+\sqrt{D_4}}{2\left(Q^2+\sum_{i=1}^n q_i^2\right)}\right\rangle; & \begin{array}{l}\text{if } D_4 > 0,\, D_3 > 0, \\ \sqrt{D_4} \ge 2Q\sum_{i=1}^n q_i^2, \\ \sqrt{D_3} < 2Q\sum_{i=1}^n q_i^2\end{array} \\ \left\langle\frac{2Q\sum_{i=1}^n q_i^2+\sqrt{D_3}}{2\left(Q^2+\sum_{i=1}^n q_i^2\right)}; \frac{2Q\sum_{i=1}^n q_i^2+\sqrt{D_4}}{2\left(Q^2+\sum_{i=1}^n q_i^2\right)}\right\rangle; & \begin{array}{l}\text{if } D_4 > 0,\, D_3 > 0, \\ \sqrt{D_4} \ge 2Q\sum_{i=1}^n q_i^2, \\ \sqrt{D_3} \ge 2Q\sum_{i=1}^n q_i^2\end{array} \end{cases} \tag{19}$$

The industry is considered to be 'concentrated' after its entry (let the production volume of the new entity in the case of 'concentrated' industry be denoted as ΔQ^C), if applied, we get the following:

$$\Delta Q^C \in \begin{cases} (0; \infty); & \text{if } D_4 < 0 \\ \left(0; \frac{2Q\sum_{i=1}^n q_i^2-\sqrt{D_4}}{2\left(Q^2+\sum_{i=1}^n q_i^2\right)}\right) \\ \quad \cup \left(\frac{2Q\sum_{i=1}^n q_i^2+\sqrt{D_4}}{2\left(Q^2+\sum_{i=1}^n q_i^2\right)}; \infty\right); & \text{if } D_4 \ge 0, \sqrt{D_4} < 2Q\sum_{i=1}^n q_i^2 \\ \left(\frac{2Q\sum_{i=1}^n q_i^2+\sqrt{D_4}}{2\left(Q^2+\sum_{i=1}^n q_i^2\right)}; \infty\right); & \text{if } D_4 \ge 0, \sqrt{D_4} \ge 2Q\sum_{i=1}^n q_i^2 \end{cases} \tag{20}$$

6 Sensitivity analysis of the Herfindahl–Hirschman index values in cases of modified concentration intervals

Consider the analysis of the sensitivity of HHI with the afore-mentioned two basic cases:

6.1 The total industry production volume is increased after the entry of the new entity

The interval of values in which the production volume of new entrants can vary without impacting on the concentration characteristics of a given industry, are going to be again derived based on Eq. (7).

From Eq. (7) ΔQ can be expressed as the considered boundary value interval for 'moderately concentrated' industry $HHI = \frac{0.9}{n} + 0.1$:

$$\Delta Q_{1,2}^{HHI=\frac{0.9}{n}+0.1} = \frac{2\left(\frac{0.9}{n}+0.1\right)Q \pm \sqrt{\left(-2\left(\frac{0.9}{n}+0.1\right)Q\right)^2 - 4\left(1-\left(\frac{0.9}{n}+0.1\right)\right)\left(\sum_{i=1}^{n} q_i^2 - \left(\frac{0.9}{n}+0.1\right)Q^2\right)}}{2\left(1-\left(\frac{0.9}{n}+0.1\right)\right)}, \tag{21}$$

Let the expression under the square root in (21) be denoted as D_5. And for concentrated industry $HHI = \frac{0.8}{n} + 0.2$:

$$\Delta Q_{1,2}^{HHI=\frac{0.8}{n}+0.2} = \frac{2\left(\frac{0.8}{n}+0.2\right)Q \pm \sqrt{\left(-2\left(\frac{0.8}{n}+0.2\right)Q\right)^2 - 4\left(1-\left(\frac{0.8}{n}+0.2\right)\right)\left(\sum_{i=1}^{n} q_i^2 - \left(\frac{0.8}{n}+0.2\right)Q^2\right)}}{2\left(1-\left(\frac{0.8}{n}+0.2\right)\right)} \tag{22}$$

Let the expression under the square root in (22) be denoted as D_6.

6.2 The total industry production volume is unchanged after the entry of the new entity

The interval of values in which the production volume of new entrants can vary without impacting on the concentration characteristics of a given industry, reflects the fact that the industry is not able to absorb additional production volume (the volume of production is unchanged), is going to be derived based on Eq. (14).

From Eq. (14) ΔQ can be expressed as the considered boundary value interval for 'moderately concentrated' industry $HHI = \frac{0.9}{n} + 0.1$:

$$\Delta Q_{1,2}^{HHI=\frac{0.9}{n}+0.1} = \frac{2Q\sum_{i=1}^{n} q_i^2 \pm \sqrt{\left(-2Q\sum_{i=1}^{n} q_i^2\right)^2 - 4\left(Q^2+\sum_{i=1}^{n} q_i^2\right)\left(Q^2\sum_{i=1}^{n} q_i^2 - \left(\frac{0.9}{n}+0.1\right)Q^4\right)}}{2\left(Q^2+\sum_{i=1}^{n} q_i^2\right)}, \tag{23}$$

Let the expression under the square root in (23) be denoted as D_7. And for 'concentrated industry' $HHI = \frac{0.8}{n} + 0.2$:

$$\Delta Q_{1,2}^{HHI=\frac{0.8}{n}+0.2} = \frac{2Q\sum_{i=1}^{n} q_i^2 \pm \sqrt{\left(-2Q\sum_{i=1}^{n} q_i^2\right)^2 - 4\left(Q^2 + \sum_{i=1}^{n} q_i^2\right)\left(Q^2\sum_{i=1}^{n} q_i^2 - \left(\frac{0.8}{n}+0.2\right)Q^4\right)}}{2\left(Q^2 + \sum_{i=1}^{n} q_i^2\right)} \tag{24}$$

Let the expression under the square root in (24) be denoted as D_8.

6.3 Empirical analysis of concentration in the Slovak insurance market

In general, it is possible to characterize the Slovak insurance market, according to the number of companies operating in the market, as competitive. The competitive environment of the insurance market is even dynamic, evolving and constantly influenced by a wide variety of factors, which is reflected in the number of insurance products that are offered by individual insurance companies. 23 insurance companies operate in the Slovak insurance market, as of June 1 2013 are listed in Table 1.

For this analysis, data about individual insurance companies were selected for 2012 (in thousands of euro) and are listed in Table 1:

- Insurance total—Technical insurance total (IT);
- Life Insurance—Technical insurance on life insurance totals for all products (LIT) Technical insurance on life insurance - Unit-Linked (LIUL) Technical insurance on life insurance - supplementary insurance (LIS);
- Non-life insurance—Technical insurance on non-life insurance totals for all products (NLIT) Technical insurance for insurance claims on land vehicles collision damage or loss waivers (NLCDW) Technical insurance for supplemental liability protection insurance against damage caused by motor vehicles (NLSLP).

Based on the data listed in Table 1, the insurance market shares and *HHI* values of the selected types of life and non-life insurance were calculated and subsequently were allocated specified characteristics (level of concentration) of the insurance market based on EU regulations and also based on the modified concentration intervals proposed by the authors. The shares appear in Table 2.

From Table 2, it is obvious that according to the EU rules, three segments of the insurance market can be described as 'concentrated' C (NLIT $-$ $0.22 > 0.2$, NLCDW $-$ $0.23 > 0.2$ and NLSLP $-$ $0.23 > 0.2$), three segments of the insurance market as 'moderately concentrated' MC (IT $-$ $0.1 \le 0.15 \le 0.2$, LIT $-$ $0.1 \le 0.12 \le 0.2$ and LIS $-$ $0.1 \le 0.14 \le 0.2$) and one segment as 'unconcentrated' U (LIUL $-$ $0.09 < 0.1$). By applying these modified concentration intervals, four segments of the insurance market can be described as 'moderately concentrated' MC (IT $-$ $0.14 \le 0.15 \le 0.23$, NLIT $-$ $0.15 \le 0.22 \le 0.24$, NLCDW $-$ $0.19 \le 0.23 \le 0.28$ and NLSLP $-$ $0.18 \le 0.23 \le 0.27$) and three segments as 'unconcentrated' U (LIT $-$ $0.12 < 0.15$, LIUL $-$ $0.09 < 0.15$ and LIS $-$ $0.14 < 0.15$).

Table 1 The amount of insurance in thousands EUR in 2012

Name of insurance company	IT	LIT	LIUL	LIS	NLIT	NLCDW	NLSLP
AEGON Životná poist'ovňa, a.s.	32,995	32,995	21,275	10,370	0	0	0
Allianz - Slovenská poist'ovňa, a. s.	583,037	244,082	59,112	42,862	338,955	88,786	75,954
Amslico poist'ovňa—Alico, a.s.	124,408	117,541	36,484	23,223	6,867	0	0
AXA životní poist'ovna	53,942	52,370	41,558	10,618	1,572	0	0
AXA poist'ovna	12,158	0	0	0	12,158	4,006	6,787
ČSOB Poist'ovňa, a.s.	78,645	53,265	29,977	3,940	25,380	4,605	8,869
D.A.S. poist'ovňa právnej ochrany, a.s.	2,492	0	0	0	2492	0	0
Deutscher Ring Lebensversicherung—AG	5,780	5,780	5,727	53	0	0	0
Deutscher Ring Sachversicherung—AG	3,776	0	0	0	3776	0	0
ERGO životná poist'ovňa, a.s.	12,149	12,149	11,694	248	0	0	0
Generali Slovensko poist'ovňa, a.s.	181,125	79,217	47,374	10,918	101,908	34,508	20,823
Groupama Garancia poist'ovňa, a.s.	6,874	456	4	58	6,419	1,560	3,450
ING Životná poist'ovňa, a.s.	78,450	78,450	37,445	20,368	0	0	0
KOMUNÁLNA poist'ovňa, a.s., VIG	165,558	106,718	17,622	4,153	58,840	18,315	30,312
KOOPERATIVA poist'ovňa, a.s., VIG	491,759	241,786	32,269	16,671	249,973	69,028	104,017
Poist'ovňa Cardif Slovakia, a.s.	15,941	4,008	0	0	11,933	0	0
Poist'ovňa Poštovej banky,a.s.	8,287	7,148	452	3,603	1,139	0	0
Poist'ovňa Slovenskej sporitel'ne, a. s., VIG	53,018	53,018	30,183	2,040	0	0	0
Union poist'ovňa,a.s.	42,322	10,540	2,415	1,647	31,782	3,964	5,263
UNIQA poist'ovňa, a.s.	102,333	31,563	11,261	4,076	70,770	26,908	17,514
Victoria - Volksbanken Poist'ovňa, a. s.	2,563	1,699	710	0	864	0	0
Wüstenrot poist'ovňa, a.s.	57,383	33,501	21,124	1,390	23,882	4,751	12,950
Slovenská kancelária poist'ovatel'ov	12	0	0	0	12	0	12
Total	2,115,007	1,166,286	406,686	156,238	948,722	256,431	285,951

Source: Slovak Insurance Association (Premiums Written 2013)

Table 2 Insurance market share of certain types of life and non-life insurance in 2012

Name of insurance company	IT	LIT	LIUL	LIS	NLIT	NLCDW	NLSLP
AEGON Životná poist'ovňa, a.s.	1.560 %	2.829 %	5.231 %	6.637 %	0.000 %	0.000 %	0.000 %
Allianz - Slovenská poist'ovňa, a.s.	27.567 %	20.928 %	14.535 %	27.434 %	35.728 %	34.624 %	26.562 %
Amslico poist'ovňa—Alico, a.s.	5.882 %	10.078 %	8.971 %	14.864 %	0.724 %	0.000 %	0.000 %
AXA životní poist'ovna	2.550 %	4.490 %	10.219 %	6.796 %	0.166 %	0.000 %	0.000 %
AXA poist'ovna	0.575 %	0.000 %	0.000 %	0.000 %	1.282 %	1.562 %	2.373 %
ČSOB Poist'ovňa, a.s.	3.718 %	4.567 %	7.371 %	2.522 %	2.675 %	1.796 %	3.102 %
D.A.S. poist'ovňa právnej ochrany, a. s.	0.118 %	0.000 %	0.000 %	0.000 %	0.263 %	0.000 %	0.000 %
Deutscher Ring Lebensversicherung—AG	0.273 %	0.496 %	1.408 %	0.034 %	0.000 %	0.000 %	0.000 %
Deutscher Ring Sachversicherung—AG	0.179 %	0.000 %	0.000 %	0.000 %	0.398 %	0.000 %	0.000 %
ERGO životná poist'ovňa, a.s.	0.574 %	1.042 %	2.875 %	0.159 %	0.000 %	0.000 %	0.000 %
Generali Slovensko poist'ovňa, a.s.	8.564 %	6.792 %	11.649 %	6.988 %	10.742 %	13.457 %	7.282 %
Groupama Garancia poist'ovňa, a.s.	0.325 %	0.039 %	0.001 %	0.037 %	0.677 %	0.608 %	1.207 %
ING Životná poist'ovňa, a.s.	3.709 %	6.726 %	9.207 %	13.037 %	0.000 %	0.000 %	0.000 %
KOMUNÁLNA poist'ovňa, a.s., VIG	7.828 %	9.150 %	4.333 %	2.658 %	6.202 %	7.142 %	10.600 %
KOOPERATIVA poist'ovňa, a.s., VIG	23.251 %	20.731 %	7.935 %	10.670 %	26.348 %	26.919 %	36.376 %
Poist'ovňa Cardif Slovakia, a.s.	0.754 %	0.344 %	0.000 %	0.000 %	1.258 %	0.000 %	0.000 %
Poist'ovňa Poštovej banky, a.s.	0.392 %	0.613 %	0.111 %	2.306 %	0.120 %	0.000 %	0.000 %
Poist'ovňa Slovenskej sporitel'ne, a. s., VIG	2.507 %	4.546 %	7.422 %	1.306 %	0.000 %	0.000 %	0.000 %
Union poist'ovňa, a. s.	2.001 %	0.904 %	0.594 %	1.054 %	3.350 %	1.546 %	1.841 %
UNIQA poist'ovňa, a. s.	4.838 %	2.706 %	2.769 %	2.609 %	7.460 %	10.493 %	6.125 %
Victoria - Volksbanken Poist'ovňa, a. s.	0.121 %	0.146 %	0.175 %	0.000 %	0.091 %	0.000 %	0.000 %
Wüstenrot poist'ovňa, a. s.	2.713 %	2.872 %	5.194 %	0.890 %	2.517 %	1.853 %	4.529 %

Table 2 continued

Name of insurance company	IT	LIT	LIUL	LIS	NLIT	NLCDW	NLSLP
Slovenská kancelária poist'ovatel'ov	0.001 %	0.000 %	0.000 %	0.000 %	0.001 %	0.000 %	0.004 %
Number of insurance companies in the market (n)	23	19	18	17	18	10	11
HHI value	0.1549	0.1232	0.0880	0.1426	0.2209	0.2277	0.2272
Characteristics of market according to EU	*MC*	*MC*	*U*	*MC*	*C*	*C*	*C*
Modified value of lower boundary for unconcentrated market $HHI = \frac{1}{n}$	0.043478	0.052632	0.055556	0.058824	0.055556	0.1	0.090909
Modified value of lower boundary for moderately concentrated market $HHI = \frac{0.9}{n} + 0.1$	0.13913	0.147368	0.15	0.152941	0.15	0.19	0.181818
Modified value of lower boundary for concentrated market $HHI = \frac{0.8}{n} + 0.2$	0.234783	0.242105	0.244444	0.247059	0.244444	0.28	0.272727
Characteristics of market according to modified boundaries	*MC*	*U*	*U*	*U*	*MC*	*MC*	*MC*

Source: Own calculations

Characteristics of market: U 'Unconcentrated' market; MC 'Moderately concentrated' market; C 'Concentrated' market

Next, we will analyze the possibility of changing the characteristics of the market after introducing a new insurance product into that segment. It is obvious that for some insurance products, in cases when product offered is interesting for customers, the increase in the total volume of insurance needs to be considered (for example LIT, LIUL, LIS). A new product entry, thus, increases the total volume insurance in this segment and it is interesting to see how much the volume of the new insurance product would have to be so that it did not change the characteristics of the segment. Calculations based on Eqs. (11), (12) and (13) are carried out in Table 3 (according to EU methodology) and in Table 4 [according to the modified concentration intervals methodology—Eqs. (21) and (22)].

For some types of insurance products, it may be assumed that the entry of a new product into the market will not increase the total volume insurance in this segment, because only a certain number of items or individuals can be insured (for example NLCDW, NLSLP). Allowable concentration intervals changes are determined based on Eqs. (18), (19) and (20) and are listed in Table 5 (according to EU methodology) and in Table 6 (according to the modified concentration intervals methodology—Eqs. (23) and (24)).

The following labeling is used in Tables 3, 4, 5 and 6:

n	Number of entities offering insurance products in a given segment,
$D_1, D_2, D_3, D_4, D_5, D_6, D_7, D_8$	Determinants in Eqs. (9), (10), (16), (17), (21), (22), (23) and (24),
Interval U	Interval, in which the value of the amount of a new insurance product can vary, so that the relevant segment can be characterized as 'unconcentrated'.
Interval MC	Interval, in which the value of the amount of a new insurance product can vary, so that the relevant segment can be characterized as 'moderately concentrated'.
Interval C	Interval, in which the value of the amount of a new insurance product can vary, so that the relevant segment can be characterized as 'concentrated'.

In Table 3, the boundaries of concentration intervals are calculated according to EU methodology, for those cases when the total insurance volume will increase after the entry of a new insurance product. Segments like NLIT, NLCDW and NLSLP can be characterized on the basis of values *HHI* as 'concentrated' (see Table 2). In Table 3 from row named '*Interval U'* it is obvious that even after the entry of a new insurance product into the segment, market cannot reach such a concentration that it can be characterized as 'unconcentrated'. If the volume of a new insurance product varies in the interval specified in Table 3 from row '*Interval MC'*, the characteristics of the segment would change from concentrated to moderately concentrated. For example segment concentration NLIT will change to moderately concentrated, if new insurance product enters the market with the volume of insurance in range $\langle 56, 357.9; 418, 003\rangle$

Table 3 Determination of concentration intervals according to the EU, in cases where the total volume of insurance in the market will increase after the new insurance product enters the market

	IT	LIT	LIUL	LIS	NLIT	NLCDW	NLSLP
n	23	19	18	17	18	10	11
D_1	$-7.05 \cdot 10^{11}$	$-5.94 \cdot 10^{10}$	$1.37 \cdot 10^{10}$	$-2.76 \cdot 10^{9}$	$-3.56 \cdot 10^{11}$	$-2.76 \cdot 10^{10}$	$-3.42 \cdot 10^{10}$
D_2	$1.36 \cdot 10^{12}$	$5.5 \cdot 10^{11}$	$8.57 \cdot 10^{10}$	$8.39 \cdot 10^{9}$	$8.37 \cdot 10^{10}$	$4.68 \cdot 10^{9}$	$5.96 \cdot 10^{9}$
$\Delta Q_1^{HHI=0.1}$	NA	NA	110,294	NA	NA	NA	NA
$\Delta Q_2^{HHI=0.1}$	NA	NA	−19,920	NA	NA	NA	NA
$\Delta Q_1^{HHI=0.2}$	1,257,990	755,824	284,655	96,315.7	418,003	106,874	119,735
$\Delta Q_2^{HHI=0.2}$	−200,487	−172,681	−81,312	−18,197	56,357.9	21,341.8	23,240.1
Interval U	$\varnothing$	$\varnothing$	$(0; 110, 294)$	$\varnothing$	$\varnothing$	$\varnothing$	$\varnothing$
Interval MC	$\langle 0; 1, 257, 990\rangle$	$\langle 0; 755, 824\rangle$	$\langle 110{,}294; 284, 655\rangle$	$\langle 0; 96, 315.7\rangle$	$\langle 56, 357.9; 418, 003\rangle$	$\langle 21, 341.8; 106, 874\rangle$	$\langle 23, 240.1; 119, 735\rangle$
Interval C	$(1, 257, 990; \infty)$	$(755, 824; \infty)$	$(284, 655; \infty)$	$(96, 315.7; \infty)$	$(0; 56, 357.9) \cup (418, 003; \infty)$	$(0; 21, 341.8) \cup (106, 874; \infty)$	$(0; 23, 240.1) \cup (119, 735; \infty)$

Source: Own calculations

Table 4 Determination of concentration intervals according to the modified concentration intervals methodology, in cases where the total volume of insurance in the market will increase after a new insurance product enters the market

	IT	LIT	LIUL	LIS	NLIT	NLCDW	NLSLP
N	23	19	18	17	18	10	11
D_5	$1.04 \cdot 10^{11}$	$2.3 \cdot 10^{11}$	$4.97 \cdot 10^{10}$	$3.14 \cdot 10^{9}$	$-1.36 \cdot 10^{11}$	$1.45 \cdot 10^{9}$	$-1.34 \cdot 10^{9}$
D_6	$2.08 \cdot 10^{12}$	$8.09 \cdot 10^{11}$	$1.18 \cdot 10^{11}$	$1.36 \cdot 10^{10}$	$2.79 \cdot 10^{11}$	$3.05 \cdot 10^{10}$	$3.52 \cdot 10^{10}$
$\Delta Q_1^{HHI=\frac{0.9}{n}+0.1}$	528,707	482,882	202,940	61,299.2	NA	83,678.1	NA
$\Delta Q_2^{HHI=\frac{0.9}{n}+0.1}$	154,932	−79,721	−59,403	−4,879.9	NA	36,622.9	NA
$\Delta Q_1^{HHI=\frac{0.8}{n}+0.2}$	1,591,316	965,977	358,618	128,828	656,530	221,036	236,129
$\Delta Q_2^{HHI=\frac{0.8}{n}+0.2}$	−293,470	−220,850	−95,468	−26,297	−42,651	−21,589	−21,666
Interval U	$(154,932; 528,707)$	$(0; 482,882)$	$(0; 202,940)$	$(0; 61,299.2)$	$\varnothing$	$(36,622.9; 83,678.1)$	$\varnothing$
Interval MC	$\langle 0; 154,932\rangle \cup \langle 528,707; 1,591,316\rangle$	$\langle 482,882; 965,977\rangle$	$\langle 202,940; 358,618\rangle$	$\langle 61,299.2; 128,828\rangle$	$\langle 0; 656,530\rangle$	$\langle 0; 36,622.9\rangle \cup \langle 83,678.1; 221,036\rangle$	$\langle 0; 236,129\rangle$
Interval C	$(1,591,316; \infty)$	$(965,977; \infty)$	$(358,618; \infty)$	$(128,828; \infty)$	$(656,530; \infty)$	$(221,036; \infty)$	$(236,129; \infty)$

Source: Own calculations

Table 5 Determination of concentration intervals according to the EU, in cases where the total volume of insurance in the market will remain the same after a new insurance product enters the market

	IT	LIT	LIUL	LIS	NLIT	NLCDW	NLSLP
n	23	19	18	17	18	10	11
D_3	$-1.41 \cdot 10^{37}$	$-1.1 \cdot 10^{35}$	$3.76 \cdot 10^{32}$	$-1.65 \cdot 10^{30}$	$-2.88 \cdot 10^{35}$	$-1.19 \cdot 10^{32}$	$-2.29 \cdot 10^{32}$
D_4	$2.72 \cdot 10^{37}$	$1.02 \cdot 10^{36}$	$2.34 \cdot 10^{33}$	$5 \cdot 10^{30}$	$6.781 \cdot 10^{34}$	$2.02 \cdot 10^{31}$	$3.98 \cdot 10^{31}$
$\Delta Q_1^{HHI=0.1}$	NA	NA	86,763.9	NA	NA	NA	NA
$\Delta Q_2^{HHI=0.1}$	NA	NA	−20,946	NA	NA	NA	NA
$\Delta Q_1^{HHI=0.2}$	788,811	458,614	167,450	59,584.1	290,160	75,434.6	84,396.3
$\Delta Q_2^{HHI=0.2}$	−221,481	−202,692	−101,632	−20,595	53,197.8	19,702.1	21,493.3
Interval U	$\varnothing$	$\varnothing$	(0; 86763.9)	$\varnothing$	$\varnothing$	$\varnothing$	$\varnothing$
Interval MC	$\langle 0; 788,811\rangle$	$\langle 0; 458,614\rangle$	$\langle 86,763.9; 167,450\rangle$	$\langle 0; 59,584.1\rangle$	$\langle 53,197.8; 290,160\rangle$	$\langle 19,702.1; 75,434.6\rangle$	$\langle 21,493.3; 84,396.3\rangle$
Interval C	$(788,811; \infty)$	$(458,614; \infty)$	$(167,450; \infty)$	$(59,584.1; \infty)$	$(0; 53,197.8) \cup (290,160; \infty)$	$(0; 19,702.1) \cup (75,434.6; \infty)$	$(0; 21,493.3) \cup (84,396.3; \infty)$

Source: Own calculations

Table 6 Determination of concentration intervals according to the modified concentration intervals methodology, in cases where the total volume of insurance in the market will remain the same after a new insurance product enters the market

	IT	LIT	LIUL	LIS	NLIT	NLCDW	NLSLP
n	23	19	18	17	18	10	11
D_7	$2.07 \cdot 10^{36}$	$4.26 \cdot 10^{35}$	$1.36 \cdot 10^{33}$	$1.87 \cdot 10^{30}$	$-1.1 \cdot 10^{35}$	$6.28 \cdot 10^{30}$	$-8.95 \cdot 10^{30}$
D_8	$4.16 \cdot 10^{37}$	$1.5 \cdot 10^{36}$	$3.22 \cdot 10^{33}$	$8.13 \cdot 10^{30}$	$2.26 \cdot 10^{35}$	$1.32 \cdot 10^{32}$	$2.35 \cdot 10^{32}$
$\Delta Q_1^{HHI=\frac{0.9}{n}+0.1}$	422,972	341,492	135,383	44,025.9	NA	63,090.5	NA
$\Delta Q_2^{HHI=\frac{0.9}{n}+0.1}$	144357	−85,570	−69,565	−5,037.2	NA	32,046.1	NA
$\Delta Q_1^{HHI=\frac{0.8}{n}+0.2}$	908,082	528,361	190,571	70,607.7	388,017	118,711	129,331
$\Delta Q_2^{HHI=\frac{0.8}{n}+0.2}$	−340,752	−272,439	−124,753	−31,619	−44,659	−23,574	−23,442
Interval U	$(144,357; 422,972)$	$(0; 341,492)$	$(0; 135,383)$	$(0; 44,025.9)$	$\varnothing$	$(32,046.1; 63,090.5)$	$\varnothing$
Interval MC	$\langle 0; 144,357\rangle \cup \langle 422,972; 908,082\rangle$	$\langle 341,492; 528,361\rangle$	$\langle 135,383; 190,571\rangle$	$\langle 44,025.9; 70,607.7\rangle$	$\langle 0; 388,017\rangle$	$\langle 0; 32,046.1\rangle \cup \langle 63,090.5; 118,711\rangle$	$\langle 0; 129,331\rangle$
Interval C	$(908,082; \infty)$	$(528,361; \infty)$	$(190,571; \infty)$	$(70,607.7; \infty)$	$(388,017; \infty)$	$(118,711; \infty)$	$(129,331; \infty)$

Source: Own calculations

thousands of EUR. If the volume of a new insurance product varies in the interval specified in row '*Interval C*' the characteristics of the segment would remain unchanged and equal to concentrated. For example segment concentration NLIT will remain the same—concentrated, if new insurance product enters the market with the volume of insurance in ranges $(0; 56, 357.9)$ or $(418, 003; \infty)$ thousands of EUR.

On the basis of the values of *HHI* listed in Table 2, the segments of market IT, LIT and LIS are characterized as moderately concentrated. According to the calculations of interval boundaries obtained (see Table 3), it is obvious that even in this case, after the entry of a new insurance product into the segment, it cannot reach such a concentration that it can be characterized as unconcentrated ('*Interval U*'). If the volume of a new insurance product varies in the interval specified in row '*Interval MC*', the characteristics of the segment would not change and would remain as moderately concentrated. For example segment concentration IT will remain the same moderately concentrated, if new insurance product enters the market with the volume of insurance in range $\langle 0; 1, 257, 990\rangle$ thousands of EUR. If the volume of a new insurance product varies in the interval specified in row '*Interval C*', the characteristics of the segment would change to concentrated. For example segment concentration IT will change to concentrated, if new insurance product enters the market with the volume of insurance in range $(1, 257, 990; \infty)$ thousands of EUR.

On the basis of values *HHI* listed in Table 2, segments of market LIUL are characterized as 'unconcentrated'. According to the interval boundaries obtained and listed in Table 3 in the row named *Interval U* $(0; 110, 294)$, it is obvious that, if the volume of a new insurance product varies in this range, the characteristics of the segment would not change. If the volume of a new insurance product varies in the range $110, 294; 284, 655$ specified in the row *Interval MC*, the characteristics of the segment LIUL would change to 'moderately concentrated'. If the volume of a new insurance product varies in the interval $(284, 655; \infty)$ specified in the row *Interval C*, the characteristics of the segment would change to 'concentrated'.

In Table 4, the boundaries of concentration intervals are calculated according to the modified concentration intervals methodology, for cases where the total insurance volume will increase after the entry of a new insurance product.

On the basis of values *HHI* listed in Table 2, market segments LIT, LIUL and LIS are characterized as 'unconcentrated' and market segments IT, NLIT, NLCDW and NLSLP as 'moderately concentrated'. The interval values listed in the rows named *Interval U*, *Interval MC* and *Interval C* determine the volumes of a new insurance product in the given market in order to change (or not to change) the characteristics of the concentration of the given market segment. The way of interpretation of numerical values from Table 4 is analogous to interpretation from Table 3.

The interpretation of results from Table 5 can be done in an analogous manner as for Table 3, except here it is for cases where the total volume of insurance in the market after the entry of a new insurance product will not change.

The interpretation of results from Table 6 can be done in an analogous manner as for Table 4, except here it is for cases where the total volume of insurance in the market after the entry of a new insurance product won't change.

7 Conclusion

In this paper we presented a modification of the levels of concentration values measured by the Herfindahl–Hirschman index and a sensitivity analysis of their change relative to the Slovak insurance market. The approach presented allows simulations of concentration changes in a given industry or given market after the entry of a new entity on market. In our consideration, we have assumed that the market is not able to absorb the additional production volume of a new entity and, therefore, its arrival in the market won't expand total production, but the production of entities already operating in a given market shall be reduced proportionately, or, in the second case, we assume that the industry absorbs the additional volume of production and total production increases after the entry of a new entity in the market. With respect to the European Union Commission methodology, the above considerations are supported by deriving relationships used to determine boundaries in which the characteristics of industry concentration ('unconcentrated', 'moderately concentrated', 'concentrated') would not change. In the same way, one can derive the equations corresponding to the US Federal Trade Commission methodology. The derived equations for sensitivity analysis can be successfully used as a tool in assessing new entrants in any industry. The use of the above analysis, which is based on the methodology of the European Commission and the US Federal Trade Commission, may be difficult for economies in which a smaller number of undertakings operate in the market. The determination of boundary intervals for assessing the level of concentration in the industry could, thus, be derived just from the lower boundary value $1/n$. Accepting an interval for the distribution $\langle 1/n, 1\rangle$ in deciles, value $0.9/n+0.1$ resp. $0.8/n+0.2$ would represent the boundary interval values. Then it would be possible to modify the classification of the level of industry concentration as follows: 'unconcentrated' industry for *HHI* values ranging in the interval $\left\langle\frac{1}{n}\,;\,\frac{0,9}{n}+0,1\right)$, 'moderately concentrated' industry for *HHI* values ranging in the interval $\left\langle\frac{0,9}{n}+0,1;\,\frac{0,8}{n}+0,2\right\rangle$ and 'concentrated industry' for *HHI* values ranging in the interval $\left(\frac{0,8}{n}+0,2;\,1\right\rangle$. A modified classification of levels of industry concentration can be considered in cases, where the number of operating undertakings in a given market is $n \geq 6$.

These assumptions were used in the empirical analysis of the Slovak insurance market and a sensitivity analysis was carried out on real data from 2012. The solutions gained from applying Eqs. (9), (10), (16), (17), (21), (22), (23) and (24) are presented in Tables 3, 4, 5 and 6. This study documents the use of a sensitivity analysis for various level of concentration based on *HHI*, when a new entity enters (insurance product) the insurance market.

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